

SMART INDIA HACKATHON 2026



Problem Statement ID: 26149

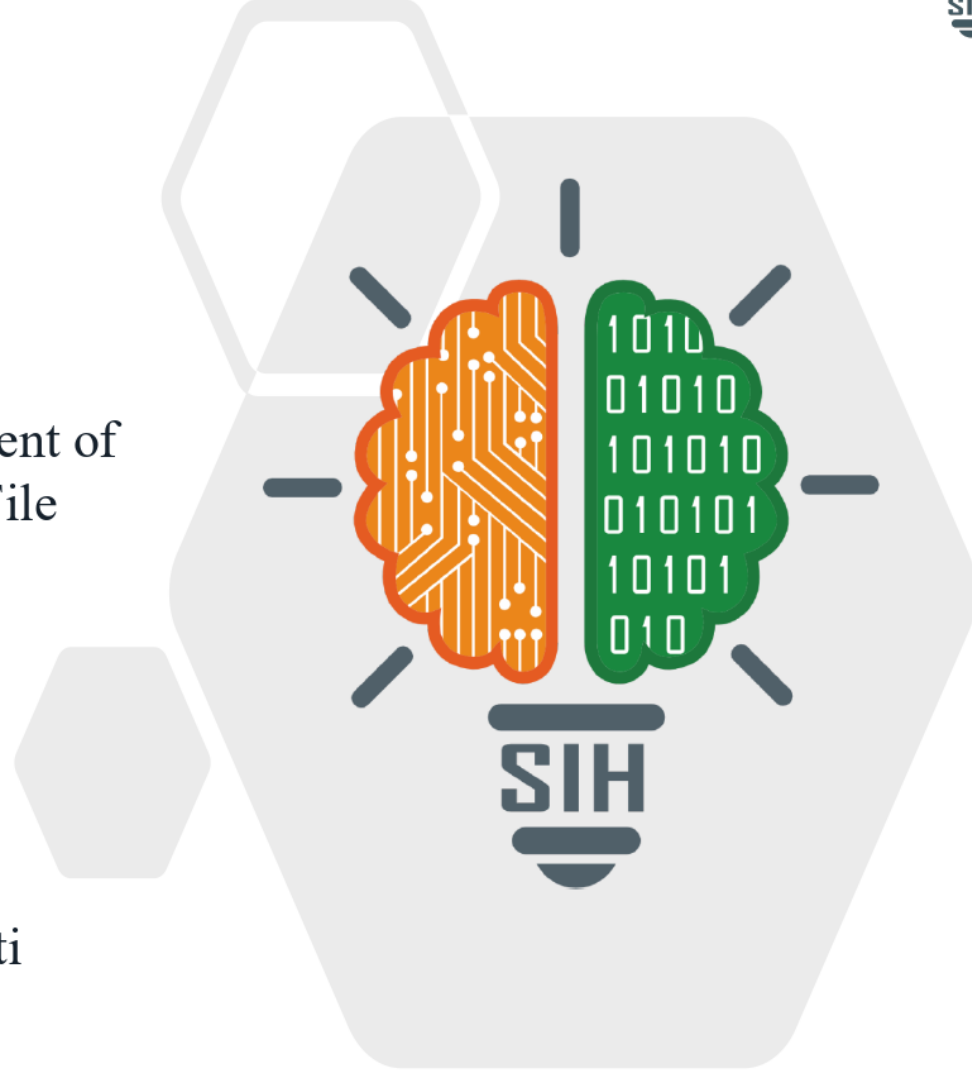
Problem Statement Title: Design and Development of an Integrated Secure Data Erasure and Advanced File Recovery Tool for Digital Forensics and Data Sanitization

Theme: Blockchain & Cybersecurity

PS Category: Software

Team ID: 148458

Team Name (Registered on portal): Data_Drishti



Sanitization that cannot be proven is indistinguishable from deletion;
recovery that cannot be trusted is indistinguishable from fabrication.
Both halves are one machine, wired to one hash-chained audit record.



GUI Dashboard Screenshot

[\(Demo link\)](#)

Innovation & Uniqueness (UVP)

- **Structural Parsing:** Walks internal chunk and marker headers to eliminate phantom and truncated carves.
- **Auditable Scoring:** Derives confidence from transparent, visible arithmetic sums (+/- weights).
- **Zero-Dependency Core:** Forensic engines run completely offline on air-gapped workstations without external packages.



SANCTUM:

One integrated core for secure erasure, deep recovery,
and a provable record of both.



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- Detailed explanation of the proposed solution
- How it addresses the problem
- Innovation and uniqueness of the solution



Proposed Solution

- Unified platform combining drive/file sanitization and file recovery into one core engine.
- Integrates 3 functional modules: Drive Eraser, File Eraser, and Signature Carver.
- Records every action in a back-linked, HMAC-sealed SHA-256 audit chain.



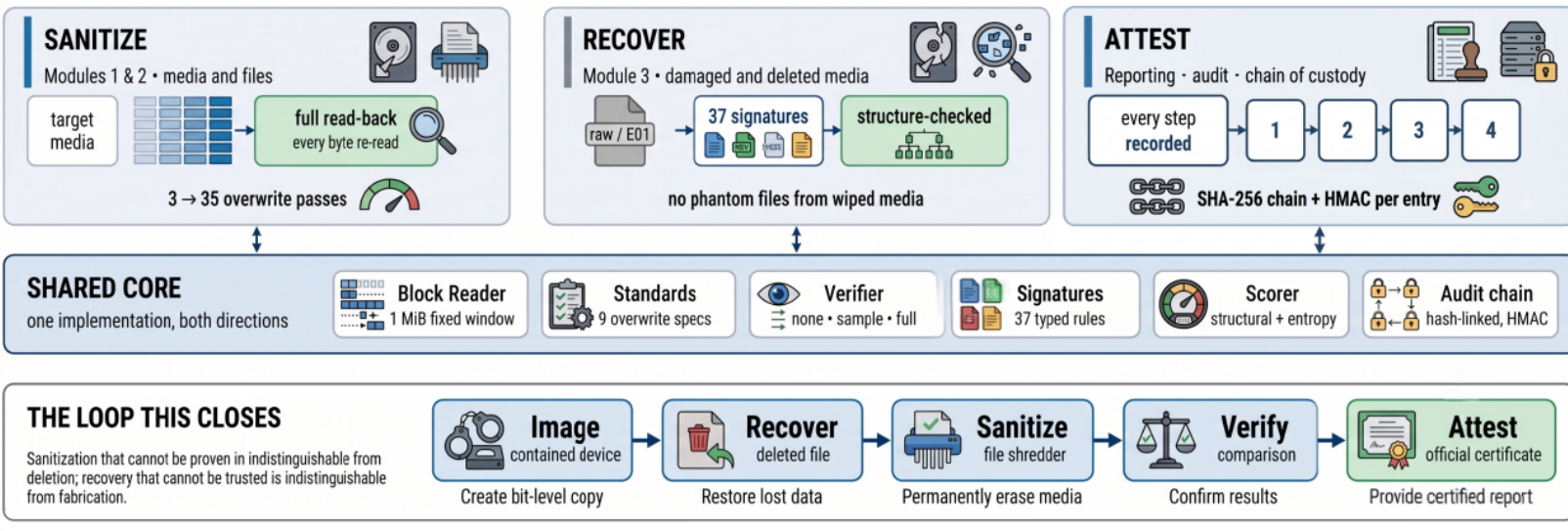
How It Addresses the Problem

- Replaces disparate tools with a single, end-to-end auditable workflow.
- Prevents catastrophic mis-wipes via a 4-gate safety lock and device-derived tokens.
- Enforces honest reporting by flagging media limitations like flash wear-leveling.



ONE INTEGRATED CORE

Sanitization and recovery are the same machine pointed in two directions – so both are wired to one hash-chained audit record.



Structure-validated recovery
not header-only carving

Auditable confidence score
every factor kept in the report

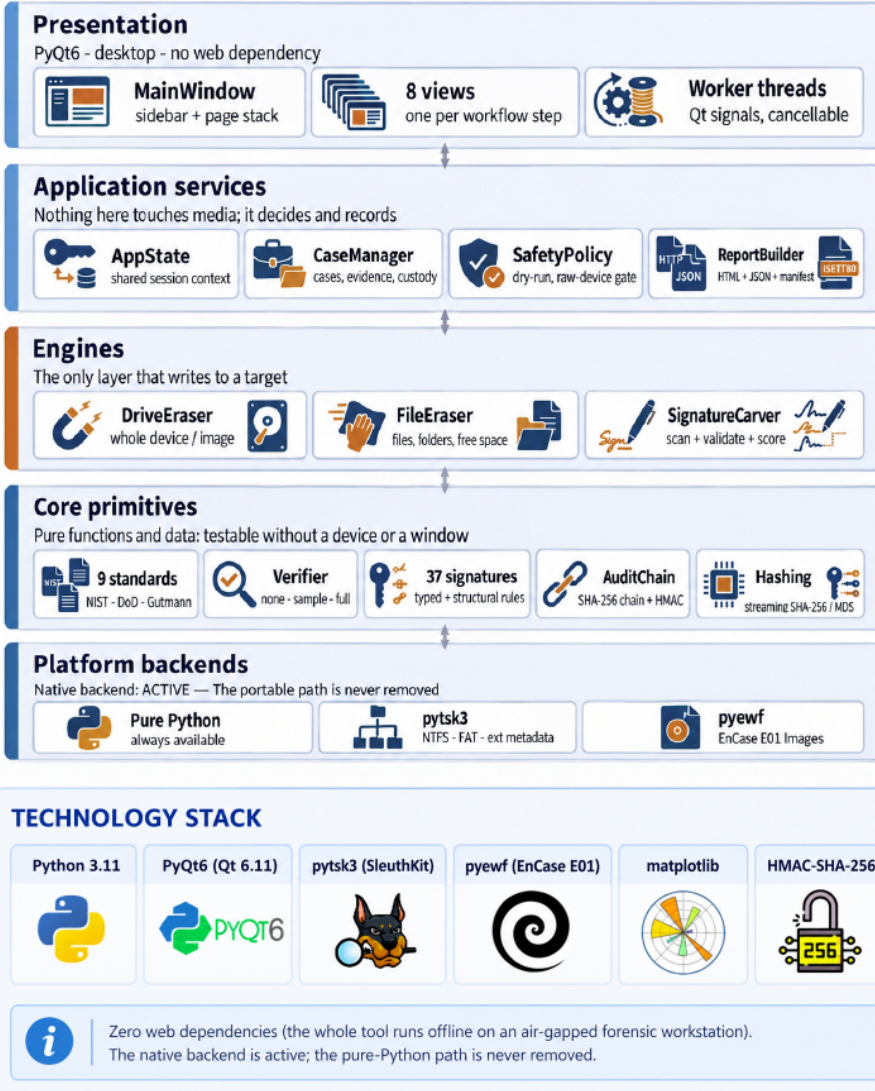
Tamper-evident audit chain
SHA-256 + HMAC per entry

Read-only on every source
recovery cannot alter evidence

37 signatures • 9 standards
E01, NTFS, FAT, ext

LAYERED ARCHITECTURE

Each layer is testable without the one above it; the GUI cannot do anything the engines cannot already do.



DATA FLOW — CARVING AND RECOVERY

Read-only on the source. Every candidate is validated before it is reported, and every accepted artefact is idigested and logged.



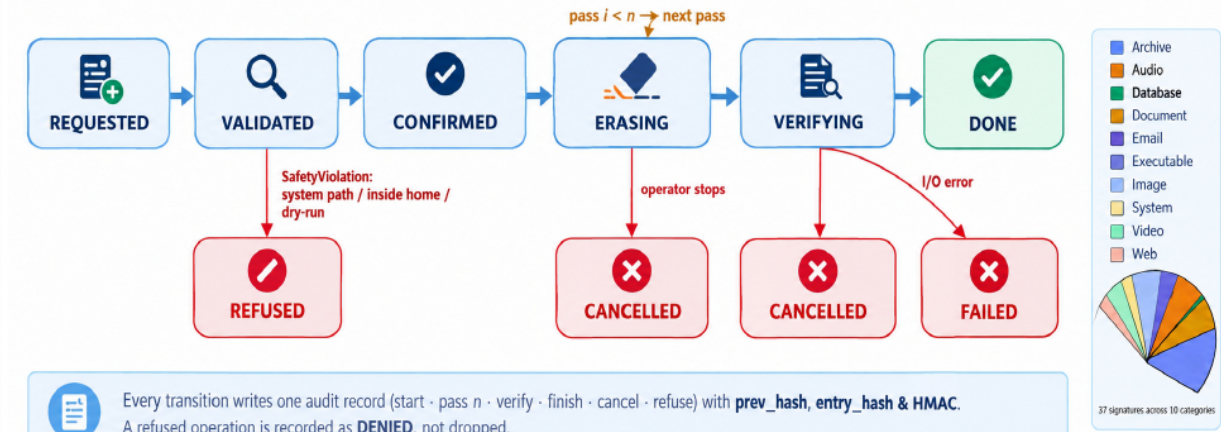
AUDIT SPINE — every stage above writes here, in order



The source handle is opened for reading and never for writing — recovery cannot alter the evidence it is recovering from.

STATE MACHINE — ONE ERASURE OPERATION

Cancellation and refusal are states, not exceptions: a batch stopped before it began raises, one stopped mid-flight returns a partial result marked cancelled.



Methodology & Process for Implementation

- Strict Layered Isolation:** UI views collect parameters only; all forensic logic lives in headless, testable engine layers.
- Multi-Gate Sanitization:** Executes 9 overwrite standards via fixed-chunk, phase-aligned block writers with mandatory read-back verification.
- Deep Structural Carving:** Employs signature matching followed by format-specific marker/chunk validation, size-boundary parsing, and additive confidence scoring.
- Immutable Audit Spine:** Every engine state transition, denial, and completed operation is appended to the back-linked HMAC audit log before returning.

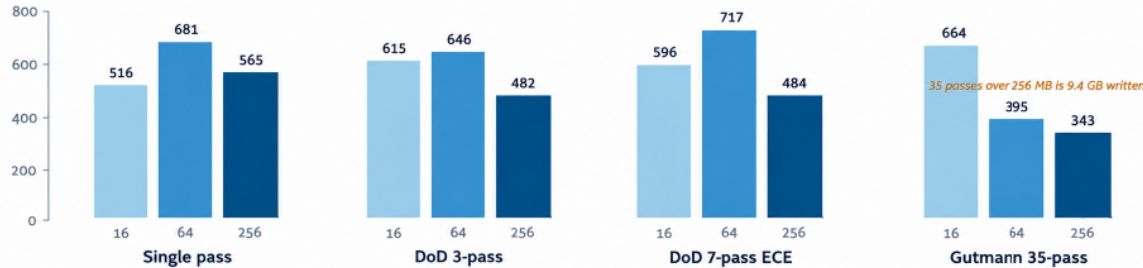
WHAT VERIFICATION COSTS

256 MB target, 7 samples per mode, interleaved in a seeded random order, each on a freshly created target. The bar is the median; the multiplier is against not verifying.



ERASURE THROUGHPUT — MB/s BY STANDARD AND SIZE

Bars are the media size: 16 MB, 64 MB, 256 MB. Throughput is flat in the size of the medium, so the cost of a wipe is the pass count and nothing else.



Gutmann is the only standard that loses throughput at scale, and it is the only one whose byte count reaches nine gigabytes.

0.90

RECALL

9 of 10 planted deleted files recovered

1.00

PRECISION

not one phantom artefact reported from sanitized or random media

11

ARTEFACTS

recovered high-confidence, each with its own SHA-256

16.04 MB

PEAK ALLOCATION

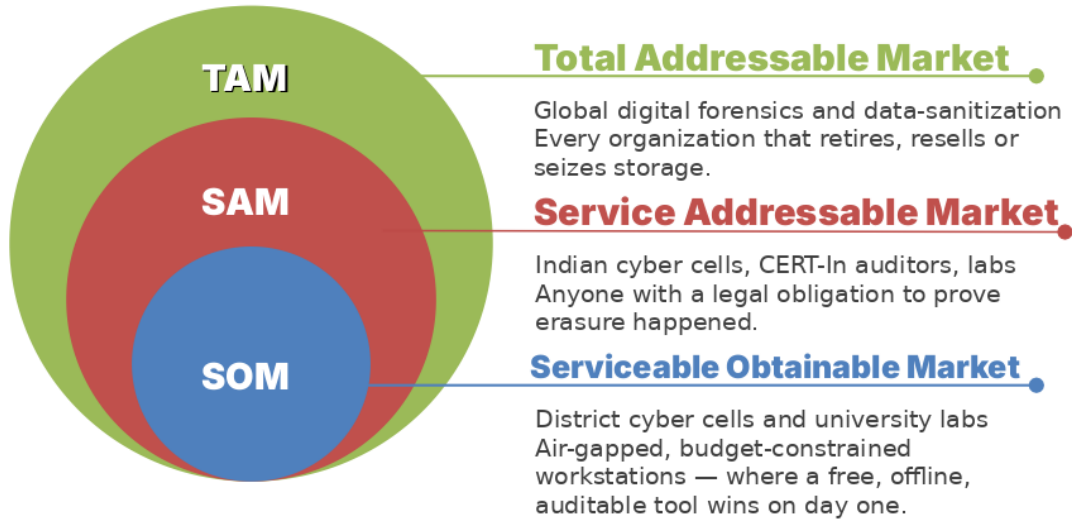
unchanged from a 32 MB image to a 128 MB one

Measured on a 16.00 MB image with a known manifest; recall and precision are SHA-256 matches, not heuristic scores. Peak allocation is measured from tracemalloc over the whole scan.

All figures from bench_final/benchmark.json on the target machine. Ratios, not absolute times, are the portable result.

Measured result	Value
Erase throughput	343–717 MB/s across 4 standards × 3 media sizes
Verification — full read-back	1.15× an unverified write
Verification — sampled	1.02× — effectively free
Carve recall	0.90 — 9 of 10 planted files
Carve precision	1.00 — no phantom artefacts
Peak allocation	16.01 MB, flat from a 32 MB to a 128 MB image
File erasure	21–22 files/s; 10× the batch costs 10.6× the time
Risk	Strategy
False positives	Structure validation + additive scoring → precision measured at 1.00
Erasing the wrong device	Triple gate: policy flag + env var + typed token; refusals audited
An unprovable wipe	Hash-chained audit + full read-back verification mode
Log tampering	Tamper-evident (SHA-256 + HMAC). Tail truncation is a documented limitation, not a hidden one
Measurement error	A reused-target bias was found and fixed; ratios published with min/max spread

ADDRESSABLE IMPACT



3 modules · 1 audited core
No second tool needed for the audit trail.

1.15× the cost of proof
Verification is cheap enough to always be on.

0 network calls
Usable on an air-gapped forensic workstation.

1.00 precision
No phantom files to explain away in court.

- **SOCIAL** — verifiable erasure protects the citizen whose data is on a retired device; verifiable recovery protects the case that depends on it.
- **ECONOMIC** — removes a paid-tool dependency for labs that cannot license commercial suites, and enables safe resale instead of scrapping.
- **ENVIRONMENTAL** — proven-safe refurbishment extends hardware life and cuts e-waste, while keeping the asset's value that physical destruction destroys.

Weakness

Desktop-only; no central case management across examiners · audit chain is tamper-evident, not tamper-proof · pure-Python path is slower on very large images.

Strength

One audit chain spans erasure and recovery · structure-validated carving suppresses false positives · fully offline · triple-gated safety on real devices · scoring is auditable factor-by-factor



Opportunity

Data-disposal and DPDP compliance duties · growing e-waste and IT-asset-disposal market · integrates with existing case management · the same core extends to network and memory captures

Threats

Commercial suites bundled into lab contracts · hardware Secure Erase / NVMe Sanitize is faster where available · SSD wear-levelling makes overwrite genuinely weaker on some media

STANDARDS AND WORKS IMPLEMENTED OR CITED

1. NIST SP 800-88 Rev. 1

Guidelines for Media Sanitization — source of the Clear / Purge / Destroy vocabulary, implemented as nist_purge.

2. DoD 5220.22-M

National Industrial Security Program Operating Manual — the 3-pass and 7-pass ECE standards.

3. P. Gutmann (1996)

Secure Deletion of Data from Magnetic and Solid-State Memory — the 35-pass scheme, and the first account of why pass count alone is not the whole story.

4. HMG Infosec Standard 5

Secure Sanitisation of Protectively Marked or Sensitive Information — hmg_enhanced.

5. VSITR

Russian state overwrite standard, 7 passes.

6. B. Schneier (1996)

Applied Cryptography — the 7-pass scheme.

7. NIST SP 800-101 Rev. 1

Guidelines on Mobile Device Forensics — integrity-recording requirements the audit chain implements.

8. NIST SP 800-86

Guide to Integrating Forensic Techniques into Incident Response — the evidence-handling workflow.

9. ISO/IEC 27037:2012

Identification, collection, acquisition and preservation of digital evidence — the chain-of-custody duty.

10. The Sleuth Kit / pytsk3

Filesystem parsing backend: NTFS, FAT, ext.

11. libewf / pyewf

EnCase E01 image format support.

12. RFC 2104

HMAC: Keyed-Hashing for Message Authentication — the per-entry MAC.

13. FIPS 180-4

Secure Hash Standard — SHA-256, used for the chain and for every artefact digest.

REPOSITORY:

[Kartik-Kashyap/sanctum-forensics: Integrated Secure Data Erasure & Forensic Recovery Platform - SIH 2026](https://github.com/Kartik-Kashyap/sanctum-forensics)

OUR RESEARCH PAPER:

<https://drive.google.com/file/d/1BuRH05yrqVKHSX37eEjgafuHKz0g5G4/view?usp=sharing>

Research Contribution & Core Findings *(find more details in our [research paper](#))*

- **Dual-Domain Evidentiary Integrity:** Formulated an integrated forensic framework where sanitization verification and carving are anchored to a unified, back-linked HMAC-SHA-256 audit spine, satisfying ISO/IEC 27037 chain-of-custody requirements.
- **Phase-Aligned Overwrite Correctness:** Identified and resolved the *pattern-phase defect* in multi-byte cyclical overwrite standards (e.g., Gutmann 3-byte sequences over 1 MiB I/O windows), preventing silent boundary desynchronization and false compliance.
- **Structural End-Detection vs. Naive Carving:** Proved that marker-chain and chunk-boundary validation prevents truncation bugs—such as EXIF embedded-thumbnail false terminations in JPEGs—achieving 1.00 precision and 0.90 recall against ground-truth FAT16 manifests.
- **Empirical Verification Costing:** Benchmarked sampled read-back verification at just 1.02× and full read-back at 1.15× over raw writes, demonstrating that evidentiary proof adds negligible wall-clock overhead while maintaining a flat 16 MB memory footprint

Research Methodology & Evolution Pipeline

